

ELBO Decomposition and the Role of Entropy

The KL divergence between the variational approximation q and the prior p decomposes as:

$$\text{KL}(q\|p) = \underbrace{-H[q]}_{\text{neg. entropy of } q} - \underbrace{\mathbb{E}_q[\log p(\beta)]}_{\text{expected log-prior}}$$

Maximising the Evidence Lower Bound (ELBO) therefore balances two forces:

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_q[\log p(\mathbf{y} \mid \beta)]}_{\text{fit the data}} - \underbrace{\text{KL}(q\|p)}_{\text{stay close to prior}}$$

KL Direction: Collapse vs. Expansion

Q: Why does $\text{KL}(q\|p)$ cause posterior variance *collapse*, while $\text{KL}(p\|q)$ causes posterior variance *expansion*?

The answer is entirely in **which distribution the expectation is taken under**:

$\text{KL}(q\|p)$ — **used in VI** —
“**mode-seeking**”

$$\mathbb{E}_q[\log q(\beta) - \log p(\beta \mid \mathbf{y})]$$

q samples its *own* regions. Where $p \approx 0$, q simply avoids that region — no penalty.

$\Rightarrow q$ **collapses onto one mode**; tails of p are ignored.

$\text{KL}(p\|q)$ — “**mean-seeking**”

$$\mathbb{E}_p[\log p(\beta \mid \mathbf{y}) - \log q(\beta)]$$

p drives the expectation. Wherever $p > 0$ and $q \approx 0$, $\log q \rightarrow -\infty$ — infinite penalty.

$\Rightarrow q$ must **cover all of p 's mass**; variance expands.